MILITARY SPECIFICATION

PRINTER, HIGH SPEED, DIGITAL DATA,
TT-624(V)/UG

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. *This specification covers the requirements for the design, test, and manufacture of a* High Speed, Digital Data Printer, TT-624(V)/UG, hereinafter referred to as the printer. The printer will generate and output hard copy of alphanumeric characters and special symbols in response to internally generated automatic signals, manually operated maintenance and operator controls, and input digital data and control signals from central processing devices, and other peripheral devices such as magnetic tape units, display units, and communication systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-B-171	Bearings, Ball, Annular, (General Purpose)
FF-B-185	Bearings, Roller, Cylindrical; And Bearings Roller, Self-Aligning
FF-B-187	Bearings, Roller Tapered
UU-P-547	Paper, Teletypewriter, Roll

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-S-3786	Switches, Rotary (Circuit Selector, Low Current Capacity) General Specification For
MIL-P-15024	Plates, Tags, And Bands For Identification Of Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-S-15291	Switches, Rotary, Snap Action And Detent/Spring Return Action, General Specification For
MIL-F-15733	Filters, And Capacitors, Radio Frequency Interference, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship *And Shore: General Specification For*
MIL-M-17059	Motors, 60-Cycle Alternating-Current, Fractional HP (Shipboard Use)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare System Command (SPAWAR 003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5815

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MIL-E-17060	Motors; Alternating-Current, Integral HP, (Shipboard Use)
MIL-M-17413	Motors; Direct Current, Integral HP, Naval Shipboard
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-M-17556	Motor, Direct Current, Fractional HP (Shipboard Use)
MIL-S-21604	Switches, Rotary, Multiple And Selector, General Specification For
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Excited Environmental And Type II - Internally)
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-242	Electronic Equipment Parts (Selected Standards)
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interface Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

MILITARY

NATIONAL SECURITY AGENCY (NSA)

NACSIM 5100A	Compromising Emanations Laboratory Test Standard Electromagnetics

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

EE-161-NA-OMI-010/E110-TT-624	Teleprinter TT-624(V)/UG, TT-624(V)1/UG, TT-624A(V)/UG, TT-624B, TT-624C(V)/UG, TT-624D(V)/UG (Basic Manual)
NAVELEX 0967-LP-544-0030	Teleprinter TT-624B(V)2/UG (188B Parallel Odd Interface)
NAVELEX 0967-LP-544-0050	Teleprinter TT-624(V)5/UG (Serial TA2 Interface)
NAVELEX 0967-LP-544-0060	Teleprinter TT-624(V)6/UG (NTDS SLOW, 8 Bit Interface)
NAVELEX 0967-LP-544-0090	Teleprinter TT-624(V)8/UG (NTDS FAST, 16 Bit Interface)
NAVELEX 0967-LP-544-0100	Teleprinter TT-624(V)9/UG (Serial ITA 2/USQ 69 Interface)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, except for associated detail specifications, specification sheets, or MS standards, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The printer shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Standard sample. A sample, representative of the type of copy to be obtained when using the printer, shall be provided as Government-furnished information.

3.2 Printer description. The printer shall be a high speed printer which provides for application as a communication terminal device. The printer shall consist of a cabinet, printer mechanics, electronic control, buffering units, and power supply. No controller shall be required for standard installation. The printer shall interface with the AN/UYK-20 digital data computer.

3.2.1 Printer external interface options. Printer external interface options shall be as specified in 3.2.1.1 through 3.2.1.8.

3.2.1.1 TT-( )(V)1/UG: standard transistor-to-transistor logic (TTL) American Standard Code for Information Interchange (ASCII) interface. The printer shall interface with the AN/UYK-20 in a ready-response mode parallel ASCII levels. The printer shall be capable of printing the sixty-four letter ASCII subset specified in TABLE I. The printer shall interface via a MS3102R-28-21PZ connector (37 pins) as specified in TABLE II.

3.2.1.2 TT( )(V)2/UG: MIL-STD-188 parallel ASCII interface. The printer shall interface with the AN/UYK-20 in a demand-response mode, MIL-STD-188 parallel ASCII levels. The printer shall be capable of printing the sixty-four letter ASCII subset specified in TABLE I. The printer shall interface via a MS3102R-36-10PZ connector (48 pins) as specified in TABLE III.

TABLE I. ASCII 64-symbol subset.

Column / b7 / b6 / b5	0 (0 0 0)	1 (0 0 1)	2 (0 1 0)	3 (0 1 1)	4 (1 0 0)	5 (1 0 1)	6 (1 1 0)	7 (1 1 1)	ROW
b4 b3 b2 b1									
0 0 0 0	NULL 1/		SPACE	0	@	P	@	P	0
0 0 0 1			!	1	A	Q	A	Q	1
0 0 1 0			"	2	B	R	B	R	2
0 0 1 1			#	3	C	S	C	S	3
0 1 0 0			$	4	D	T	D	T	4
0 1 0 1			%	5	E	U	E	U	5
0 1 1 0			&	6	F	V	F	V	6
0 1 1 1	BELL 1/		(apes)	7	G	W	G	W	7
1 0 0 0			(	8	H	X	H	X	8
1 0 0 1			)	9	I	Y	I	Y	9
1 0 1 0	LF 1/	◇ 2/	*	:	J	Z	J	Z	10
1 0 1 1			+	;	K	[	K	[	11
1 1 0 0	FF 1/		,	<	L	\	L	\	12
1 1 0 1	CR 1/		-	VS	M	]	M	]	13
1 1 1 0	SO 1/		.	>	N	♡ 3/	N	♡ 3/	14
1 1 1 1	SI 1/		/	?	O	•	O	•	15

1/ Nonprinting control characters
2/ Is the twenty-sixth character, in lieu of a space, in the printed set, indicates parity error
3/ Replaces the specified in MIL-STD-188 (row 14) and indicates an illegal character

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TABLE II. TT-624(V1) pin-outs cable connector MS3106F28-21-SZ.

Connector	Pin No.	Function
P2	G 1	CHAR DEMAND
P2	B 2	CHAR DEMAND RET.
P2	F 3	CHAR STROBE
P2	E 4	CHAR STROBE RET.
P2	H 5	BIT 1
P2	J 6	BIT 1 RET.
P2	T 7	BIT 2
P2	b 8	BIT 2 RET.
P2	U 9	BIT 3
P2	c 10	BIT 3 RET.
P2	V 11	BIT 4
P2	d 12	BIT 4 RET.
P2	W 13	BIT 5
P2	e 14	BIT 5 RET.
P2	X 15	BIT 6
P2	f 16	BIT 6 RET.
P2	h 17	BIT 7
P2	g 18	BIT 7 RET.
P2	j 19	BIT 8 (PARITY BIT OR BITP)
P2	r 20	BIT 8 RET. (PARITY OR BITP)
P2	S 21	M CLEAR
P2	a 22	M CLEAR RET.
P2	k 23	PFD (PAPER FEED)
P2	m 24	PFD (PAPER FEED) RET.
P2	A, C, D, R, P, N, M, K, L, n, p, s,	CABLE SHIELDS (CABINET GROUND CONNECT POINTS)
P2	z	UNUSED PIN

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TABLE III. TT-624(V2) pin-outs cable connector MS3106F36-10-SZ.

Connector	Pin	No.	Function
P2	k	1	CHAR DEMAND
P2	b	2	CHAR DEMAND RET.
P2	E	3	CHAR STROBE
P2	A	4	CHAR STROBE RET.
P2	g	5	READY
P2	g	6	READY RET.
P2	J	7	BIT 1
P2	C	8	BIT 1 RET.
P2	M	9	BIT 2
P2	F	10	BIT 2 RET.
P2	H	11	BIT 3
P2	0	12	BIT 3 RET.
P2	R	13	BIT 4
P2	K	14	BIT 4 RET.
P2	N	15	BIT 5
P2	T	16	BIT 5 RET.

3.2.1.3 TT-( )(V)5/UG: serial baudot interface. The printer shall interface with AN/UYK-20 in a demand response mode, MIL-STD-188 serial baudot level. The printer shall be capable of printing the sixty-four letter ASCII subset specified in TABLE I. The printer shall interface via a MS3102R-28-21PZ connector (37 pins) as specified in TABLE IV.

TABLE IV. TT-624(V5) pin-outs cable connector MS3106F28-21-SZ.

Connector	No.	Function
P2 Pin G	1	DEMAND
P2 Pin B	2	DEMAND RET.
P2 Pin H	3	DATA
P2 Pin J	4	DATA RET.

3.2.1.4 TT-( )(V)6/UG: 8-bit Naval Tactical Data System (NTDS) slow interface. The printer shall interface with the AN/UYK-20 in a demand response mode, NTDS slow 8-bit parallel format. The printer shall be capable of printing the sixty-four letter ASCII subset specified in TABLE I. The printer shall interface via a MS3102R-36-10PZ connector (48 pins) as specified in TABLE V.

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TABLE V. TT-624(V6) pin-outs cable connector MS3106F36-10-SZ.

Connector	Pin	No.	Function
P2	n	1	EXT FNCT ACK
P2	e	2	EXT FNCT ACK RET.
P2	p	3	EXT RDY STATUS
P2	v	4	EXT RDY STATUS RET.
P2	m	5	EXT INT ENA
P2	t	6	EXT INT ENA RET.
P2	s	7	BIT 8
P2	z	8	BIT 8 RET.
P2	Q	9	BIT 7
P2	X	10	BIT 7 RET.
P2	d	11	BIT 6
P2	V	12	BIT 6 RET.
P2	N	13	BIT 5
P2	T	14	BIT 5 RET.
P2	R	15	BIT 4
P2	K	16	BIT 4 RET.
P2	H	17	BIT 3
P2	O	18	BIT 3 RET.
P2	M	19	BIT 2
P2	F	20	BIT 2 RET.
P2	J	21	BIT 1
P2	C	22	BIT 1 RET.
P2	q	23	EXT INT REQ
P2	q	24	EXT INT REQ RET.
P2	J	25	EXT INT ACK
P2	r	26	EXT INT ACK RET.
P2	E	27	OUT DAT ACK
P2	A	28	OUT DAT ACK RET.
P2	k	29	OUT DAT REQ
P2	b	30	OUT DAT REQ RET.

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3.2.1.5 TT-( )(V)8/UG: 16-bit NTDS fast interface. The printer shall interface with the AN/UYK-20 in a demand-response mode; NTDS fast 11-bit parallel will perform a form feed upon receipt of a FIGURES-SPACE-FIGURES character sequence in lieu of the upper case H character. The printer shall be capable of printing the sixty-four letter ASCII subset specified in TABLE I. The printer shall interface via a MS3102R-36-10PZ connector (48 pins) as specified in TABLE VI.

3.2.1.6 TT-( )(V)9/UG serial ITA 2/USQ 69 interface. The printer shall interface with the AN/UYK-20 through the AN/USQ-69 Data Terminal in a multidrop (daisy chain) configuration so that the printer shall print data received in a serial asynchronous mode directly from the AN/USQ-69. So that the printer can be used in an additional capacity, it shall be operator-selectable to receive the print data directly from the AN/UYK-20 in serial ITA 2 mode as specified in 3.2.1.3. The printer shall interface via a MS81511/01EF01P1 and MS81511/01EF01P2 connector (85 pins) as specified in TABLE VII.

TABLE VI. TT-624(V8) pin-outs cable connector MS3106F36-10-SZ.

Connector	Pin	No.	Function
P2	k	1	ODR
P2	b	2	ODR RET.
P2	E	3	ODA
P2	A	4	ODA RET.
P2	n	5	EFA
P2	e	6	EFA RET.
P2	p	7	ERS
P2	v	8	ERS RET.
P2	D	9	BIT 1
P2	L	10	BIT 1 RET.
P2	P	11	BIT 2
P2	W	12	BIT 2 RET.
P2	U	13	BIT 3
P2	c	14	BIT 3 RET.
P2	Y	15	BIT 4
P2	f	16	BIT 4 RET.
P2	a	17	BIT 5
P2	h	18	BIT 5 RET.
P2	s	19	BIT 6
P2	x	20	BIT 6 RET.
P2	u	21	BIT 7
P2	y	22	BIT 7 RET.
P2	w	23	BIT 8
P2	z	24	BIT 8 RET.
P2	J	25	BIT 9

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TABLE VI. TT-624(V8) pin-outs cable connector MS3106F36-10-SZ. - (Continued)

Connector	Pin	No.	Function
P2	C	26	BIT 9 RET.
P2	M	27	BIT 10
P2	F	28	BIT 10 RET.
P2	H	29	BIT 11
P2	O	30	BIT 11 RET.
P2	R	31	BIT 12
P2	K	32	BIT 12 RET.
P2	N	33	BIT 13
P2	T	34	BIT 13 RET.
P2	d	35	BIT 14
P2	V	36	BIT 14 RET.
P2	Q	37	BIT 15
P2	X	38	BIT 15 RET.
P2	S	39	BIT 16
P2	Z	40	BIT 16 RET.
P2	m	41	EIE
P2	t	42	EIE RET.
P2	j	43	EIA
P2	r	44	EIA RET.
P2	q	45	EIR
P2	g	46	EIR RET.

TABLE VII. TT-624(V9) pin-outs cable connector MS81511/56FF01-S2 (for mating with connector P3) and MS81511/56FF01-S1 (for mating with connector P2). (Continued)

Connector	Pin	No.	Function
P3	20	1	RCV DATA (OUT OF 9/UG) 1/
P3	28	2	B-LINE
P3	11	3	XMIT DATA (INTO 9/UG)
P3	85	4	SIGNAL GROUND
P3	38	5	A-LINE
P3	83	6	C-LINE
P3	27	7	D-LINE

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TABLE VII. TT-624(V9) pin-outs cable connector MS81511/56FF01-S2 (for mating with connector P3) and MS81511/56FF01-S1 (for mating with connector P2). (Continued)

Connector	Pin No.	Function
P3	48 | 8	E-LINE
P3	1-10, 12-19, 21-26, 29-37, 39-47, 49-82, 84	UNUSED PIN LOCATIONS
P2	3 | 1	RCV DATA (OUT OF 9/UG) 1/
P2	16 | 2	B-LINE
P2	6 | 3	XMIT DATA (INTO 9/UG)
P2	7 | 4	SIGNAL GROUND
P2	30 | 5	A-LINE
P2	24 | 6	C-LINE
P2	40 | 7	D-LINE
P2	17 | 8	E-LINE
P2	23	K-LINE (NO CONNECT)
P2	1, 2, 4, 5, 8-15, 18-22, 25-29, 31-39, 41-85	UNUSED PIN LOCATIONS
P2	6	DATA (INTO 9/UG) 2/
P2	16	DEMAND
P2	7	SIGNAL GROUND
P2	3, 30, 24, 40, 17, 23	RESERVED PIN LOCATIONS (NOT USED WITH 9/UG IN ITA MODE)
P2	1, 2, 4, 5, 8-15, 18-22, 25-29, 31-39, 41-85	UNUSED PIN LOCATIONS

1/ The 9/UG can be internally configured to operate in one of two modes. The 9/UG has two input/output connectors to allow daisy chaining (or busing) of peripherals when used in the USQ mode. Only one connector is utilized when the 9/UG is used in the USQ mode and the 9/UG is the only unit in the chain or the last unit in the chain. Both connect points are functionally identical. The differences are the connector shell keying and signal pin-out.

2/ The 9/UG can be internally configured to operate in one of two modes. The 9/UG has two connectors to allow daisy chaining (or busing) of peripherals when used in the USQ mode. Only one connector is utilized when the 9/UG is used in the ITA mode. Both connect points are functionally identical. The differences are the connector shell keying and signal pin-out. This is based on the use of P2 of the 9/UG as the connect point for ITA mode operation.

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3.2.1.7 External interfaces. All printer interfaces specified in 3.2.1 through 3.2.1.6 shall interface with external equipment as specified in a through f:

a. TT-( )(V)1/UG in accordance with EE161-NA-010/E110-TT-624 (Basic Manual)
b. TT-( )(V)2/UG in accordance with NAVELEX 0967-LP-544-0030, Supplement 0031 through Supplement 0034
c. TT-( )(V)5/UG in accordance with NAVELEX 0967-LP-544-0050, Supplement 0051 through Supplement 0056
d. TT-( )(V)6/UG in accordance with NAVELEX 0967-LP-544-0060, Supplement 0061 through Supplement 0065
e. TT-( )(V)8/UG in accordance with NAVELEX 0967-LP-544-0090, Supplement 0091 through Supplement 0093
f. TT-( )(V)9/UG in accordance with NAVELEX 0967-LP-544-0100

3.2.1.8 Connector. One associated mating connector shall be furnished with each equipment configuration.

3.2.2 Printed character codes. The printed character codes shall be as specified in TABLE I. Any code received which is not specified in TABLE I shall be interpreted as an illegal character and shall cause the heart symbol, (octal ACSII 136 or 176) to be printed. The printer shall be equipped with odd parity detection. When the printer is in parity detect mode and a parity error is detected in an otherwise valid character, the diamond symbol (octal code 032) shall be printed in its place.

3.2.3 Control character codes. The control character shall be as specified in TABLE I. Any code received which is not specified in TABLE I shall be interpreted as an illegal character and shall cause the heart symbol (octal 136 or 176) to be printed.

3.2.3.1 NULL code. The NULL code shall be received by the printer, checked for parity, and ignored.

3.2.3.2 BELL code. The BELL code shall cause the audible alarm to sound for a 0.25-second. Additional BELL codes received within a 0.25-second shall extend the audible alarm to a 1-second period.

3.2.3.3 Line feed (LF) code. The LF code shall cause the previously stored data to be printed and the paper to advance on line, without restoring the left margin.

3.2.3.4 Form feed (FF) code. The FF code function shall cause the previously stored data to be printed, and shall initiate paper movement to the next complete increment of 66 lines and restore the left margin. During operation of FF additional motion commands sent to the printer will be ignored.

3.2.3.5 Carriage return (CR) code. The CR code shall cause the previously stored data to be printed, advance the paper one line, and restore the left margin. The printer shall automatically perform the CR sequence after the detection of a printable character corresponding to the end of line at the seventy-second or eightieth column (depending on 72- or 80-column switch setting). LF followed by CR or FF will cause the paper to advance without restoring the left margin. The printer shall ignore any CR or LF (or combination thereof) in the next one or two successive character locations immediately following a previously received CR code.

3.2.3.6 Shift out (SO) code. The SO code shall cause all successive printed characters to be emphasized (underlined).

3.2.3.7 Shift in (SI) code. The SI code shall terminate the SO code by ending the character emphasis.

3.2.3.8 Start of special sequence (SS) code. The SS code shall cause the diamond symbol (octal 032) to be printed.

3.3 Parts, materials, and processes. Parts, materials, and processes shall be as specified in 3.3.1 through 3.3.2.

3.3.1 Parts. Parts shall be as specified in 3.3.1.1 through 3.3.1.9.

3.3.1.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.1.2 Bearings. Bearings shall be in accordance with MIL-STD-454, Requirement 6. Ball bearings shall be in accordance with FF-B-171, Grade 00 where possible. Roller bearings shall be in accordance with FF-B-185 or FF-B-187. Sleeve bearings shall be oil-impregnated phosphor-bronze and shall be used only when approved by the procuring activity.

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3.3.1.3 Fasteners. Fasteners and latches shall be provided which shall be consistent with the degree of enclosure and the frequency of use required. Fasteners and latches shall be a quick connect-disconnect type whenever practicable.

3.3.1.4 Filters. Radio frequency interference filters shall conform to the requirements of MIL-F-15733.

3.3.1.5 Motors. Motors shall conform to Service A of MIL-M-17059, MIL-M-17060, MIL-M-17413, or MIL-M-17556.

3.3.1.6 Printed circuit boards (PCBs). PCBs within each module or line repair item (LRI) (see 6.4.1) shall use plug-in construction to facilitate ease of removal and insertion.

3.3.1.7 Printed wiring. Printed wiring shall be in accordance with MIL-STD-454, Requirement 17. Coatings shall be Type PUR of MIL-I-46058.

3.3.1.8 Springs. Springs shall be in accordance with MIL-STD-454, Requirement 41, except hammer support springs.

3.3.1.9 Switches. Rotary switches shall be in accordance with MIL-S-3786, MIL-S-15291, or MIL-S-21604.

3.3.2 Exterior finish. The exterior of the cabinet shall be coated with enamel in accordance with MIL-E-15090, Type III, Class 2.

3.4 Characteristics. Printer characteristics shall be as specified in 3.4.1 through 3.4.1.21.

3.4.1 Performance characteristics. Unless otherwise specified herein, the printer and major components shall conform to the performance characteristics specified herein under the standard conditions specified in 3.4.1.1 and the environmental conditions specified in 3.7.

3.4.1.1 Character repertoire. The alphanumeric and symbol repertoire of the printer shall be ASCII in accordance with the 64-Symbol Subset (Page Printer Application) table of MIL-STD-188, (Appendix C), except as specified in TABLE I.

3.4.1.2 Character size and style. The font style shall be open Gothic. The printed character size shall be 0.110 inches (in.) (2.79 millimeters (mm)) high and 0.075 in. (1.91 mm) wide.

3.4.1.3 Character spacing. Horizontal spacing shall be 10 characters per inch.

3.4.1.4 Legibility. All characters shall be precise and easily legible from a normal reading distance (about 14 in. (355.6 mm)). No part of a character shall drop out, nor shall there be any fading to a point where fading is readily noticed (provided that a good ribbon is in use).

3.4.1.5 Print speed. The print rate for printing the full 64 character ASCII subset shall be not less than 500 characters per second (CPS) with a throughput rate of 375 lines per minute (80 character lines including CR and LF) using a sliding test pattern.

3.4.1.6 Paper line advance time. The maximum time allowed for paper motion shall be 27 milliseconds or less from start of motion command to paper stop.

3.4.1.7 Paper slew speed. The paper slew speed shall be 2 in. (50.8 mm) per second, or faster.

3.4.1.8 Print margins. A switch shall be provided on the maintenance panel to select an automatic carriage return at either 72 columns or 80 columns of printing. The 72-column selection is used when the fanfolded paper width is less than 9.25 in. (234.9 mm). The tractors shall normally be left-justified, but both left and right tractors shall be adjustable to accommodate varying paper widths for various print-field configurations. Only 8.438 in. (214.31 mm) wide roll-type paper shall be used in the friction feed system.

3.4.1.9 Printer paper mechanism. The printer shall provide the capability to use, interchangeably, roll or fanfolded paper mechanisms. Each equipment shall be provided with one roll paper mechanism and one fanfolded paper mechanism.

3.4.1.9.1 Roll paper mechanism. The friction feed mechanism of the printer shall be capable of printing on the 5-in. (127.0-mm) diameter roll-types of paper specified in a and b:

a. UU-P-547 teletype paper, from single-part (carbonless) up to four-part (original plus three carbons and copies), FSN 7520-223-7966
b. Four-part National Cash Register paper, or equivalent

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3.4.1.9.2 Fanfolded paper mechanism. In the sprocket feed mechanism the printer shall be capable of using any standard fanfolded paper 8.5 in. (215.9 mm) through 9.875 in. (250.83 mm) wide. The printer will accept 15-pound (lb) carbon-interleaved paper (original with five carbons and copies).

3.4.1.10 Paper loading time. Paper loading shall require less than 90 seconds by a radioman second class (RM-2) skill level.

3.4.1.11 Paper feed interchangeability time. The printer shall be capable of being changed from the friction fed to the sprocket fed version (or vice versa) in less than 10 minutes by an RM-2 or RM-3 skill level.

3.4.1.12 Internal paper capacity. The printer shall have internal storage for one 5-in. (127.0-mm) teletype roll of paper or a 4.5-in. (114.3-mm) stack of fanfolded paper.

3.4.1.12.1 External paper feed. The printer shall be capable of accepting fanfolded paper directly from an external supply box.

3.4.1.13 Contrast of printed data. The contrast of original and copies shall be similar to that of a good typewriter using the same type of paper (see 3.1.2).

3.4.1.14 Line length. The printer shall have a normal capacity of 80 characters per line, printing any character in any position. The printer shall be capable of operating at 72 characters per line (see 3.4.1.8).

3.4.1.15 Line spacing. The vertical mean-line-to-mean-line spacing shall be 0.167 in. (4.24 mm) ±0.010 in. (0.25 mm) (noncumulative error). At this spacing the normal page content is 66 lines, in the fanfolded version.

3.4.1.16 Printed line visibility. Printed characters shall be visible through the viewing window after eight line feeds, that is, about 1.3 in. (33.02 mm).

3.4.1.17 Top of form. The printer shall be capable of leaving seven unprinted lines around the perforations at the top of fanfolded paper. The normal setting shall be one line above and six lines below the fold. A control shall be provided to disable this feature on roll paper.

3.4.1.18 Printed data removal. The printer shall be capable of operating continuously, without adverse effect, while a printed message is torn off, with the paper in motion.

3.4.1.19 Horizontal registration. Spacing from character to character shall be 0.100 in. (2.544 mm) ±0.005 in. (0.13 mm) with a maximum accumulated tolerance of ±0.010 in. (0.25 mm) over 80 columns (8 in. (203.2 mm)).

3.4.1.20 Vertical registration. Any character in a printed line shall be within 0.010 in. (0.25 mm) of its true designated vertical position (see 6.4.4).

3.4.1.21 Ribbon change time. The ribbon, if required, shall be capable of being changed in less than 3 minutes by a RM-2 or RM-3 skill level.

3.4.2 Physical characteristics. The printer shall be quiet, reliable, and rugged. The printer shall be capable of producing single or multiple copies. The paper drive system shall be capable of being converted from sprocket feed to friction feed or vice versa, allowing use of teletype paper rolls or edge-punched fanfolded paper stacks. The printer shall have the physical characteristics specified in 3.4.2.1 through 3.4.2.4.

3.4.2.1 Weight. The printer shall weigh not more than 290 lbs.

3.4.2.2 Dimensions. The dimensions of the printer shall not exceed 22 in. ±0.13 in. (558.8 mm ±3.3 mm) high, 22.25 in. ±0.13 in. (565.15 mm ±3.3 mm) deep (without basket or connector), and 26.5 in. ±0.13 in. (673.1 mm ±3.3 mm) wide.

3.4.2.3 Paper exit. The printed message shall exist at the top of the printer cabinet. A message tear-bar shall be provided. It shall be possible to tear messages from the printer while the paper is moving, without jamming.

3.4.2.4 Interface capability. The printer shall have provisions for substituting additional buffering capacity and a data compression feature to allow operation as a data communication terminal device. The printer shall provide the capability to substitute the interfaces specified in a through f:

a. Standard TTL ASCII Interface
b. MIL-STD-188 parallel ASCII Interface

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c. Serial ITA2 Interface
d. NTDS SLOW, 8 Bit Interface
e. NTDS FAST, 16 Bit Interface
f. Serial ITA2/USQ 69 Interface

3.5 Design. Design requirements shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Maintenance design. Test measurement and diagnostic equipment test provisions for the equipment shall be in accordance with MIL-STD-454, Requirement 32, except as specified herein.

3.5.1.1 Accessibility. Principal assemblies, subassemblies, and parts of the printer shall be accessible for operation, maintenance, and adjustment from the top or front of the printer. All test points, and adjustments, required for organizational maintenance, shall be accessible from the top or front of the printer while the equipment is operational, and without requiring removal of cables or connectors. The use of extender boards is permitted to allow access to test points and adjustments on printed circuit cards. The equipment shall be designed such that special purpose electronic test equipment (SPETE) may be connected to the equipment without disconnecting external equipment cables.

3.5.1.2 Doors and access panels. Doors and access panels shall either be removable or have provisions to secure them in the open position. Hinged access panels shall be employed whenever the frequency of removal warrants.

3.5.1.3 General purpose electronic test equipment (GPETE). GPETE for use at all levels of maintenance shall be selected from the standard and substitute standards listed in MIL-STD-1364.

3.5.1.4 SPETE. SPETE shall, if required, be as specified in a and b:

a. The use of SPETE at the organizational level of maintenance shall be limited to:
   1. Self-test card
   2. Extender card assembly
b. The use of SPETE at the depot level of maintenance shall include the SPETE at the organizational level of maintenance plus on-line exerciser that emulates AN/UYK-20 and AN/USC-69.

3.5.1.5 Special tools. Special tools, if required, shall be supplied in accordance with the contract (see 6.4.3). Special purpose tools shall be as defined in the contract.

3.5.2 Mechanical design. The equipment shall conform to the applicable mechanical design requirements of MIL-E-16400 and as specified in 3.5.2.1 through 3.5.2.4.19.

3.5.2.1 Enclosure. The printer shall be dripproof (15 degrees) in accordance with MIL-STD-108.

3.5.2.2 Mounting. The printer shall have provisions to enable bolting to a stand or table. The mounting shall be accomplished without mechanical disassembly.

3.5.2.3 Noise. The printer shall conform to the noise requirements of MIL-STD-740 for Grade A, Type 3 equipment.

3.5.2.4 Controls, indicators, and alarms. Controls, indicators, and alarms shall be provided to control equipment modes of operation and to notify operators visually and audibly of equipment status.

3.5.2.4.1 Equipment electrical power ON-OFF switch. A switch or circuit breaker for disconnecting equipment from all electrical power systems shall be mounted on the equipment front panel and the function clearly labeled. The switch or circuit breaker shall break all power conductors of the power circuit.

3.5.2.4.1.1 Power indicator lamp. A green power indicator lamp shall be mounted near the switch to indicate when the equipment is energized.

3.5.2.4.2 Illumination control. A dimmer shall be provided to control indicator brightness.

3.5.2.4.3 Form feed switch. The form feed switch shall cause all stored data to be printed and shall cause paper to advance continuously until the top of form sensor is detected. The incoming data shall take priority and shall cause the switch function to be delayed until completion of loading of one character. The printer shall inhibit the data request line until the form feed is executed.

3.5.2.4.4 READY or OFF-LINE indicators. The READY indicator, when illuminated, shall indicate that the printer is able to accept and print data from the data source. The OFF-LINE indicator, when illuminated, shall indicate that the printer is off-line.

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3.5.2.4.5 ON-LINE or OFF-LINE switch. The printer shall have a push-push type switch to select the ON-LINE or OFF-LINE modes. The printer shall be in an operation mode when the switch is in the ON-LINE position and all other READY conditions are satisfied. The switch in the OFF-LINE position shall inhibit communication to the control device. After a fault condition is corrected, a transition from the OFF-LINE to the ON-LINE is required before the printer can be ready again.

3.5.2.4.6 PAPER FAULT or PAPER LOW indicators. The PAPER FAULT indicator shall be illuminated whenever paper continuity has been broken. The PAPER LOW indicator shall indicate that the paper supply is nearing the end of the roll in the friction feed configuration. This shall correspond with the red strips normally found on a paper roll, indicating about 10 feet (ft) to 15 ft remaining.

3.5.2.4.7 Circuit breakers. Circuit breakers shall be provided for circuit protection. No fuses shall be used. A visual indicator shall be provided to indicate the state of each circuit breaker. On the main alternating current (AC) power circuit breaker, both sides of the power line shall be open-circuited when the circuit breaker is off.

3.5.2.4.8 Audible alarm. An audible alarm shall be provided to notify operator personnel whenever a PAPER FAULT or PAPER LOW condition exists. In addition, the alarm shall be activated upon receipt of a BELL code character from the data source. The time duration for the audible alarm shall be as specified in a through c:

a. Paper Fault or Low: 10 seconds
b. BELL code - signal BELL code causes the alarm to sound for 0.25 second
c. Additional BELL codes received within the first 0.25-second period shall extend the alarm to 1 second.

3.5.2.4.9 Alarm volume control. A volume control shall be provided to vary the sound level of the alarm.

3.5.2.4.10 Copy control and indicator. A control device shall be provided to compensate for the variations in paper thickness between single and multiple part paper. An integral indicator shall indicate the correct copy setting.

3.5.2.4.11 Master clear switch. A switch shall be provided to initialize all control logic and clear the buffer.

3.5.2.4.12 Parity error detect or ignore switch. The printer shall be equipped with a switch to activate or deactivate the parity checking logic.

3.5.2.4.13 Print inhibit switch. A switch shall be provided to prevent hammer motion and override the paper interlock.

3.5.2.4.14 Skip over in or out switch. The printer shall be equipped with a switch to activate or deactivate the fanfold paper perforation skip-over feature as specified in 3.4.1.8.

3.5.2.4.15 72-column or 80-column switch. A switch shall be provided to select either 72-column or 80-column printing (see 3.4.1.7 and 3.4.1.8). The carriage shall automatically return after printing the selected 72 columns or 80 columns.

3.5.2.4.16 Knobs. Knobs shall be in accordance with MIL-STD-242-12H.

3.5.2.4.17 Lifting points. Lifting points shall be provided if required.

3.5.2.4.18 Viewing window. The printer shall have a viewing window to enable operator personnel to monitor the printed material.

3.5.2.4.19 Elapsed time meter (ETM). An ETM shall be provided which shall allow for visual indicator of either print time, power-on time, or on-line time. The ETM shall consist of a single standard size printed wiring board located within the electronic bay and will receive power and signals from the main wire wrap back plane. The time reading will be in hours, up to 9999. The ETM shall be visible when the electronic bay door is open.

3.5.3 Thermal design. The equipment, if required, shall be air-cooled with ventilation openings and filters as specified in the Ventilation openings paragraph of MIL-E-16400.

3.6 External power. The printer shall be completely self-contained, requiring only a single source of AC power. The printer shall operate from 115 volt (V) ±10 percent, 60 hertz (Hz), ±5 percent, single-phase power. Maximum continuous power consumption shall not exceed 350 watts with a power factor of greater than 80 percent.

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3.7 Environmental conditions. The printer shall conform to all performance requirements specified herein under the environmental conditions specified in 3.7.1 through 3.7.4.

3.7.1 Temperature. The printer shall conform to all performance requirements when subjected to the Range 4 temperature requirements of MIL-E-16400.

3.7.2 Humidity. The printer shall conform to the humidity requirements of MIL-E-16400 without paper in the machine. With paper in the machine, the maximum value of the relative humidity shall be 85 percent.

3.7.3 Shock. The printer shall be capable of withstanding the shock requirements of MIL-S-901, Grade A, Class I, Lightweight and up with Type A testing. The printer shall conform to all performance requirements after impact. If an intermittent failure occurs during shock, the equipment shall be returned to operating condition within 90 seconds. The printer shall be mounted in the normal shelf-mount configuration.

3.7.4 Vibration. The printer shall be capable of withstanding vibration requirements of MIL-STD-167-1 for Type I equipment.

3.8 TEMPEST. The printer shall be designed to conform to the requirements of NACSIM 5100A.

3.9 Electromagnetic environmental effects. The equipment shall be self-compatible and shall operate compatibly in the intended operating electromagnetic environment. The printer shall conform to the emission and susceptibility requirements of MIL-STD-461 for Class A4 equipment, except that the test limit for RE01 shall be as shown in FIGURE 1 and RS03 shall be conducted with a minimum of 10 volts per meter.

3.10 Identification and marking. An identification plate conforming to MIL-P-15024 shall be attached to the outside of the unit.

3.10.1 Weight marking. Equipment weighing in excess of 35 lbs shall have the weight clearly marked on the external surface of the equipment and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied.

3.11 Safety criteria. Equipment design shall conform to the Safety criteria paragraph of MIL-E-16400, except as specified in 3.11.1 and 3.11.2.

3.11.1 Power switch. When the power is in the OFF position, the printer shall be disconnected from both sides of the input power line.

3.11.2 Leakage current. The leakage current from the enclosure shall not exceed 5 milliamperes (mA).

3.12 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 750 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 250 hours. Failure shall be any part defect or equipment misalignment which prevents the equipment from conforming to the performance requirements specified herein.

3.13 Maintainability. The printer shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 35 minutes when repair is accomplished at the organizational level of maintenance by the replacement of subassemblies (PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts), except for the power supply transformer, hammer bank, power supply capacitors, and cable assemblies; these items are considered to be beyond the organizational level capability. Repair time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout.

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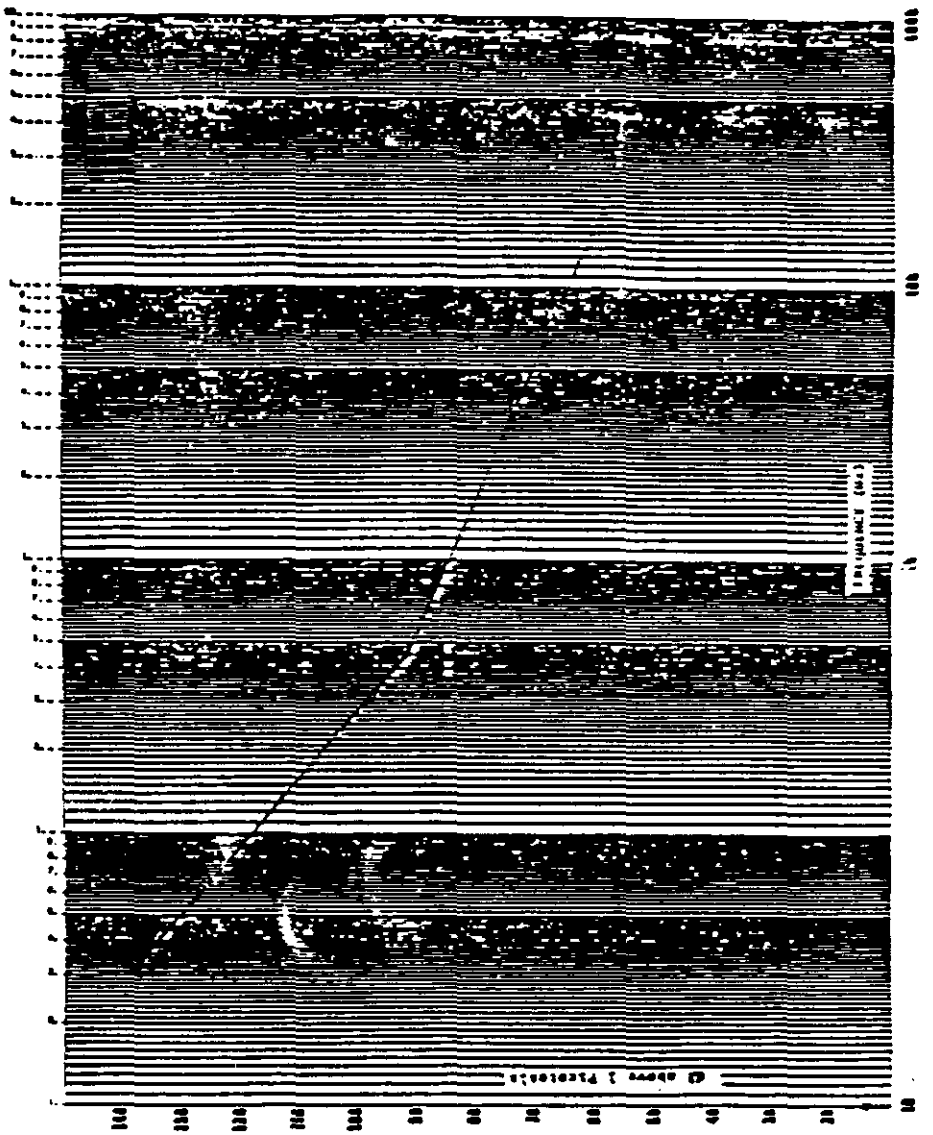


FIGURE 1. Limit for RE01.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.4 Standard test conditions. The conditions specified in a through e shall be used to measure performance characteristics:

a.	Temperature:	25 (±5° Celsius (C))(room ambient)
b.	Altitude:	Normal ground
c.	Vibration:	None
d.	Relative humidity:	Room ambient
e.	Input operating voltage:	Adjusted to 115 volts alternating current (VAC)

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C)(see 4.4.3)
c. Safety verification (see 4.7.9)
d. Inspection of packaging (see 4.11)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE VIII.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be performed on each printer offered for delivery. Production inspection shall comprise examinations and tests which will prove the workmanship and reveal degradation and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustments, and tests which detect hidden defects of materials. Production inspection shall include, but not be limited to, examinations and tests specified in Group A of TABLE VIII.

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4.4.2 Production control inspection (Group B). Production control inspection shall be performed on a sampling basis as specified in 4.4.2.1. Production control inspection shall be in addition to the production inspection specified in 4.4.1, and shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation, and tests to determine deviations in the processes and equipment required to fabricate the printer. The tests shall be performed on the complete printer as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE VIII.

4.4.2.1 Sampling for production control inspection (Group B). Printers submitted for production control inspection shall be selected by the contractor under the supervision of the Government quality assurance representative (QAR) and shall be representative of current production. Production control inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Production control inspection shall be performed on printers that have passed the production inspection specified in 4.4.1. The acceptable quality level shall be 6.5 percent defective for each of Group B examinations and tests, and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

TABLE VIII. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.1, 3.3.2, 3.4.1.12, 3.5.2, 3.5.2.2, 3.10, 3.11	4.7.1	X	X		
Operating	3.2.2, 3.2.3, 3.4.1.1, 3.4.1.4, 3.4.1.5, 3.4.1.8, 3.4.1.9, 3.4.1.11, 3.4.1.13, 3.4.1.16, 3.4.1.18 through 3.4.1.20, 3.4.2.3, 3.5.1.1, 3.5.2.4	4.7.2	X	X		
Interchangeability	3.1	4.7.3	X			
Leakage current	3.11.2	4.7.10	X		X	
Weights and dimensions	3.4.2.1, 3.4.2.2	4.7	X		X	
Steady-state voltage and frequency	3.1	4.7.5	X		X	
Transient voltage	3.1	4.7.5	X			
Transient frequency	3.1	4.7.5	X			
Power interruption	3.1	4.7.5	X			
Enclosure	3.5.2.1	4.7	X			
Temperature	3.7.1	4.7	X			X
Humidity	3.7.2	4.7	X			X
Power	3.6	4.7	X		X	
EMI emission and susceptibility and systems self-compatibility	3.9	4.7.8	X			
Controls and control circuits	3.1, 3.4.1.17, 3.5.2.4, 3.11.1	4.7, 4.7.4	X		X	
Shock	3.1, 3.7.3	4.7	X			
Vibration	3.1, 3.7.4	4.7	X			X
Inclination	3.1	4.7	X			
Noise	3.5.2.3	4.7	X			
Salt fog	3.1	4.7.6	X			
Magnetic	3.1	4.7	X			
TEMPEST	3.8	4.7.7	X			
Safety	3.11	4.7.9	X			
Reliability	3.12	4.8	X			
Maintainability	3.13	4.9	X			
General workmanship	3.1	4.10.1	X	X		
Workmanship screen	3.14	4.10.2	X	X		

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4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be accomplished on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the printer as a whole; tests of the effects of changes of environment (such as, extremes of temperature and humidity); and tests of the effects of vibration. Environmental inspection shall include the examinations and tests specified in Group C of TABLE VIII and shall be performed on units which have been subjected to, and have passed production inspection.

4.4.3.1 Sampling for environmental inspection. One printer shall be selected from each successive 125 printers, or fraction thereof, produced. The first sample printer shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4.2).

4.4.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed production control (Group B) or environmental (Group C) inspections or both, may be accepted on the contract provided the units are resubjected to, and pass, the production inspection after repair of all damage.

4.5 Performance check during test. When operation of equipment is required during test exposure, the operational checks shall be of sufficient duration or repeated at appropriate intervals to ensure a record of comprehensive comparative data with that recorded at factory ambient.

4.6 Failure criteria. Deterioration, corrosion, change of tolerance limits of any internal or external parts, or deviations from recorded performance data beyond allowed limits that in any manner could prevent the equipment from complying with operational service or maintenance requirements shall provide reason to consider the test item as having failed to withstand the conditions of test. Leakage or discoloration of impregnating compounds shall be considered damaged and cause for rejection.

4.7 Test methods. The examinations and tests specified in TABLE VIII shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.7.1 through 4.7.10.2.

4.7.1 Surface examination. The equipment shall be examined to determine compliance with the requirements specified in a through c:

a. Workmanship, assembly and fit, mechanical safety, and marking
b. Materials, parts, and finish
c. Treatment for prevention of corrosion

The examination shall be performed without disassembling the equipment in a manner that the performance, durability, or appearance would be affected.

4.7.2 Operating test. For each principal mode of operation, measurements of equipment performance shall be performed under standard test conditions (see 3.4.1) in order to determine compliance with the performance characteristics specified in a through h:

a. Character repertoire (see 3.4.1.1)
b. Legibility (see 3.4.1.4)
c. Print speed (see 3.4.1.5)
d. Print margins (see 3.4.1.8)
e. Printer paper mechanism (see 3.4.1.9)
f. Paper feed interchangeability time (see 3.4.1.11)
g. Contrast of printed data (see 3.4.1.13)
h. Controls, indicators, and alarms (see 3.5.2.4)

Factory acceptance test procedures shall be performed on each equipment before it is delivered. If a failure occurs during this test, the equipment shall be repaired and recycled through the complete factory

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acceptance test procedures until all testing requirements are satisfactorily performed and completed. The first production unit under test each month shall demonstrate the printer's capability to print on four-part roll paper (see 3.4.1.9.1) and six-part fanfold paper (see 3.4.1.9.2).

4.7.3 Interchangeability verification. The printer shall be tested to determine compliance with the performance requirements specified herein after a complete change of interchangeable assemblies and assemblies.

4.7.4 Controls and control circuitry. The equipment shall be functionally operated to determine compliance with the requirements of 3.1, 3.4.1.17, 3.5.2.4, and 3.11.1.

4.7.5 Electrical tests. Electrical tests shall be performed in accordance with MIL-E-16400 with the exceptions specified in a through c:

a. Transient voltage. Superimposed transient voltage shall be +18 percent of nominal voltage.
b. Transient frequency. Equipment shall be operating at 5 percent above or below nominal frequency before the additional 3 percent increase or decrease is imposed.
c. Power interruption. The interruption cycle consisting of three individual interruptions shall be repeated a total of four times. When the equipment under test has more than one significant operating mode, the power interruption shall be performed in each of the modes.

4.7.6 Salt fog. Finishes and coatings on parts, and frame and enclosure structures as finally assembled for use, shall be subjected to the salt fog test specified in MIL-STD-810, Method 509.2. Sample corner structures and any other critical sections may be used for the test. The test shall not be applied to the complete equipment.

4.7.7 TEMPEST. The printer shall be tested by the contractor to demonstrate compliance with NACSIM 5100A as specified in 3.8.

4.7.8 Electromagnetic emission and susceptibility tests. The requirements of 3.9 shall be verified using the applicable test methods of MIL-STD-462.

4.7.9 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.11.

4.7.10 Leakage current test. Leakage current shall be measured at 115 V, 60 Hz.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE PRINTER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.7.10.1 Printer connections. After power removal, each printer directly connected to an external power source and units deriving power from the printer shall be placed on an insulated surface. All safety ground conductors between the printer and units deriving power from the printer shall be intact. The safety ground conductor between the printer and the source power shall be opened during test. OBSERVE WARNING STATEMENT. The printer shall be connected as shown in FIGURE 2 for single-phase source power.

4.7.10.2 Measurement. Leakage current shall be measured on the printer in its normal operating configuration. Printer controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 2. The open safety conductor shall be reconnected immediately after the test is completed.

4.8 Reliability testing. A sample of three equipments selected from the initial production units shall be subjected to reliability qualification testing in accordance with Test Plan XVIC of MIL-STD-781. Environmental and operational duty cycles shall be as specified in 4.8.1 and 4.8.3. Reliability testing shall be successfully completed prior to delivery of any production equipments under the contract.

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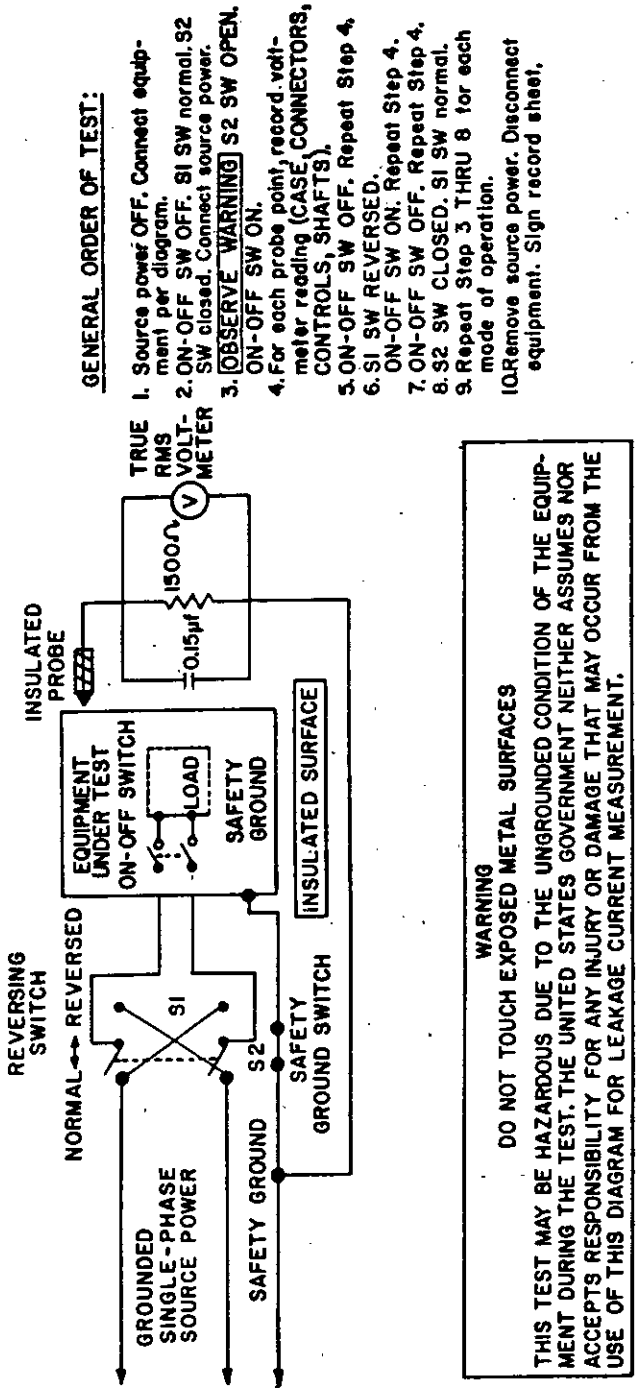


FIGURE 2. Single-phase test diagram for leakage current measurement.

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4.8.1 Temperature cycling. Temperature cycling between 0°C (+3°C, -0°C) and +50°C (+0°C, -3°C) shall be performed throughout the test with the constraints specified in a and b:

a. Time duration at the stabilized high and low temperatures shall be approximately 6 hours.
b. During transition between temperature extremes, the average rate of temperature change shall be not less than 3°C per minute.

4.8.2 Vibration. A 4-Hz to 50-Hz to 4-Hz vibration sweep shall be performed throughout the reliability test with vibratory displacement (table amplitude) as specified in the Vibratory displacement of environmental vibration table of MIL-STD-167-1. The duration of each complete sweep shall be approximately 15 minutes.

4.8.3 Operating duty cycle. Equipment shall be operating continuously during the reliability test. Input data shall be provided to the equipment which will simulate the intended printer application and which will result in a minimum on-line (PRINT mode) duty cycle of 30 percent. During unattended periods, the print impact simulator may be used to conserve paper; however, at least once daily, roll or fanfold paper shall be loaded and performance measurements taken to verify continued proper operation.

4.8.4 Input power. Input power shall be 115 VAC, 60 Hz nominal; the voltage shall be changed daily in the sequence specified in a through c:

a. Nominal (115 VAC)
b. Nominal +10 percent
c. Nominal -10 percent

4.8.5 Rejected equipment. In the event equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for rejection and propose a plan for correction of the deficiencies. When the corrections have been approved by the procuring activity and retrofitted in the test units, the reliability qualification test shall be reperformed as specified herein.

4.9 Maintainability demonstration test. Conformance to the maintainability M_{ct} requirements of 3.13 shall be verified by a maintainability demonstration in accordance with MIL-STD-471, Method 9, with a consumer's risk, β, of 90 percent. The demonstration shall consist of 30 corrective maintenance tasks. Each corrective maintenance task shall consist of repair accomplished by a subassembly or part, except for those items specified in 3.13 as being beyond the organizational level capability.

4.9.1 Selection of corrective maintenance tasks. For the maintainability demonstration, 60 candidate organizational level corrective maintenance tasks shall be determined in accordance with Appendix A of MIL-STD-471. The information specified in a through c shall be made available for each candidate task:

a. Designation of specific failed subassembly and failed part within the subassembly or failed chassis-mounted part
b. Failure mode
c. Means of introducing failure (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative shall use the 60 candidate tasks as a guide to determine a sample of 30 tasks for the demonstration.

4.9.2 Technical documentation. Technical documentation shall be limited to the equipment technical manual.

4.9.3 Rejection. If a reject decision is reached, the procuring activity shall be immediately notified.

4.9.4 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to delivery.

4.10 Workmanship. Workmanship shall be as specified in 4.10.1 through 4.10.2.2.

4.10.1 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements specified in 3.1. Each solder connection and associated wiring or leads shall be visually examined.

4.10.2 Workmanship screen. Vibration and temperature cycling shall be performed on each printer. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All hardware, including cables and connectors, shall be exposed to vibration.

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4.10.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 3. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth not greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

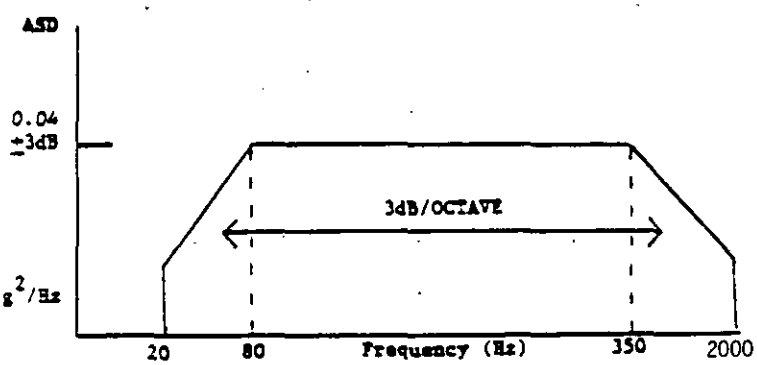


FIGURE 3. Random vibration curve.

4.10.2.2 Temperature cycling. Each printer shall be subjected to 10 cycles of the temperature curve shown in FIGURE 4. The temperature rate of exchange shall be not less than 5°C per minute. Printer power shall be turned on and off at the indicated times. The printer shall be positioned for maximum exposure to the changing temperature. Where performance measures are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the printer reaches 80 percent of the chamber temperature. When failures occur, the printer shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.11 Inspection of packaging. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The printer covered by this specification is intended for use in the Navy modular automated communication system for shipboard and shore-base use.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)
d. Levels of preservation, packaging, packing, and marking required (see 5.1)

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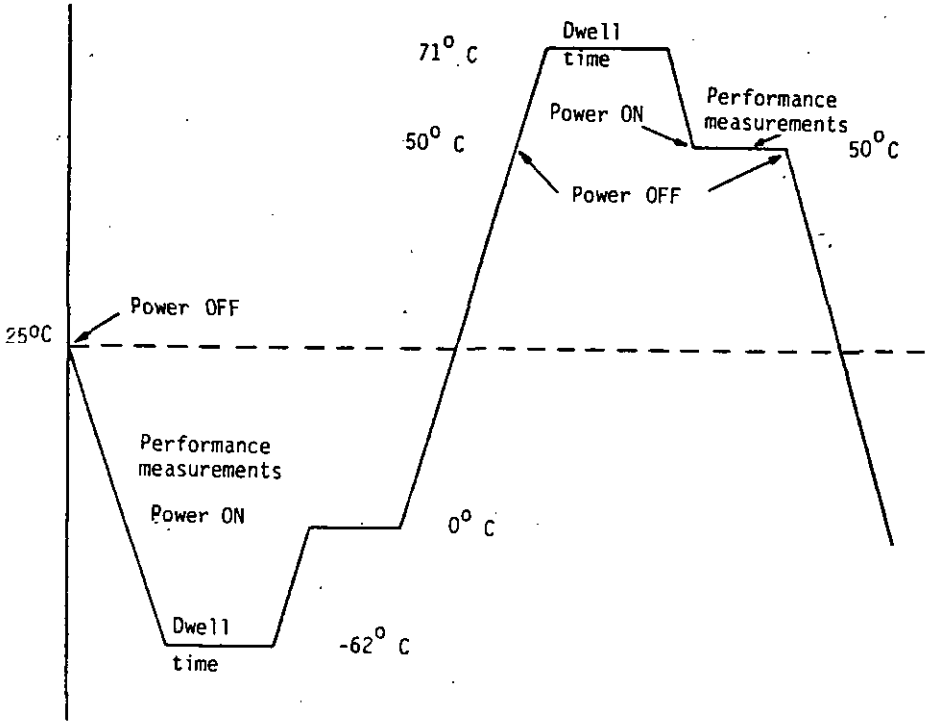


FIGURE 4. One cycle of temperature curve (not to scale).

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6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.4.

6.4.1 LRI. A LRI is that item which has been designated to be replaced at the organizational level of maintenance (module, PCB, and chassis-mounted electrical or mechanical part).

6.4.2 Failure. The term failure refers to any malfunction or parameter deviation that prevents the equipment from performing to full capability in all modes, in the specified operational environment, to the operating parameter levels specified herein.

6.4.3 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Service.)

6.4.4 True position. The distance from a line drawn parallel to a printed line and located equidistant between a feature of the highest character (such as, the top or bottom of an E) and the equivalent feature of the lowest character.

6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.5.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the environmental inspection.

6.6 Subject terms (key word) listing.

Control character codes
Electromagnetic environmental effects (E^3)
General purpose electronic test equipment (GPETE)
Intended use: shipboard and shore-based
Printed character codes
Printer, High Speed, Digital Data, TT-624(V)/UG
Special purpose electronic test equipment (SPETE)
TEMPEST
TT-624(V)/UG, Printer, High Speed, Digital Data

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NAVY-EC

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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